

BHARATIYA ANTARIKSH HACKATHON 2026

APEX-ORACLE

Physics-aware, explainable AI for detecting exoplanets in noisy TESS light curves

Problem Statement AI-enabled Detection of Exoplanets from Noisy Astronomical Light Curves

Team [YOUR TEAM NAME] **Team Leader** [YOUR NAME] (solo entry)

THE TEAM

A solo entry, full-stack ownership



[YOUR NAME]

Solo participant - ML engineering + astro-informatics

One person, every hat in the pipeline:

Data & detrending

Transformer modeling

Bayesian physics

False-positive vetting

Explainability (XAI)

Live demo & report

Why APEX-ORACLE is different

1

Different from existing ideas

Most entries are black-box CNNs that only flag a brightness dip. APEX-ORACLE also classifies the signal, returns physical parameters with confidence, and shows visual reasons for every call.

2

How it solves the problem

GP detrending preserves shallow transits, a multimodal model plus centroid vetting rejects blended false positives, and a differentiable transit model recovers parameters - even single transits.

3

Unique selling point

Detection + classification + Bayesian parameters + explainability in one validated pipeline, proven by injection-recovery tests. Accurate and fully transparent - never a black box.

CAPABILITIES

What the solution does



Transit-safe detrending

GP + Savitzky-Golay removes noise, keeps 100 ppm dips



U-vs-V disentanglement

Attention learns transit vs binary geometry



False-positive vetting

Centroid / difference imaging kills blends



Explainable visuals

Phase-folded curves with attention overlays



4-class signal typing

Transit, eclipsing binary, blend, or starspot



Differentiable parameters

Period, depth, duration with calibrated bounds



Monotransit recovery

Finds single-transit, long-period planets



Honest validation

Injection-recovery completeness & SNR

Nine-stage detection pipeline

1 Ingestion

lightkurve, astroquery

2 Detrending

wotan, celerite2

3 Candidate + fold

transitleastsquares

4 Classification

Transformer + XGBoost

5 Parameter estimation

diff. batman + sbi

6 Vetting

centroid, triceratops

7 Explainability

attention, matplotlib

8 Validation

injection-recovery

9 Live demo

Streamlit dashboard

MOCK-UP (OPTIONAL)

Live inspector dashboard

APEX-ORACLE - Exoplanet Inspector

TIC 307210830

Analyze

Phase-folded light curve + attention



TRANSIT confidence 96%

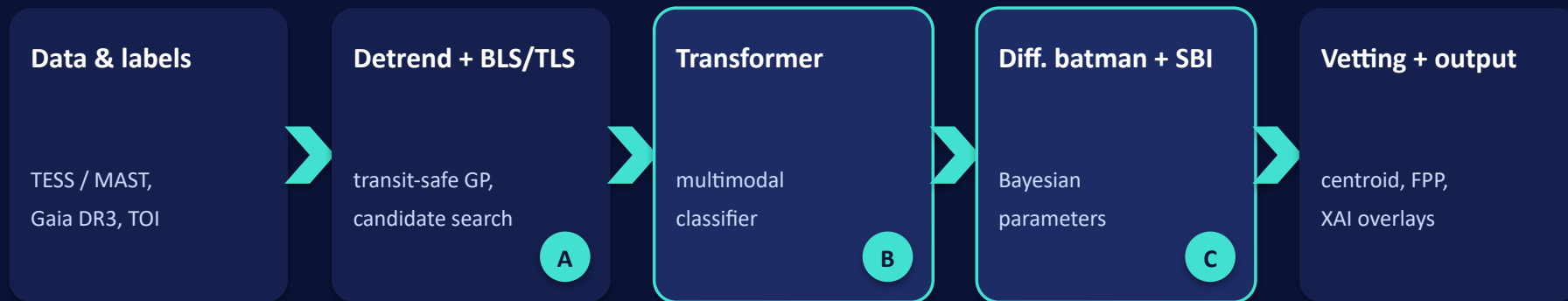
Period: **3.41 d** $\pm$ 0.02

Depth: **910 ppm** $\pm$ 40

Duration: **2.7 h** $\pm$ 0.1

Vetting: **on-target**, FPP 0.4%

APEX-ORACLE hybrid pipeline



Reproducibility & infra: uv - Hydra / OmegaConf - Weights & Biases - PyTorch Lightning - injection-recovery harness

A = fast front-end . B = detection spine . C = parameter & confidence head

TECHNOLOGY STACK

Technologies used

Data & astronomy

lightkurve - astropy - astroquery
transitleastsquares - wotan

Detrending & GP

wotan - celerite2 - gpytorch

Deep learning

PyTorch - Lightning - timm
Transformers - XGBoost

Physics & Bayesian

batman - sbi - dynesty - emcee

Vetting & XAI

triceratops - attention maps
matplotlib - plotly

Infra & delivery

uv - Hydra - Weights & Biases
Streamlit

Language: Python 3.11 . Compute: GPU for training, CPU-capable inference

ESTIMATED COST (OPTIONAL)

Near-zero, fully reproducible

Data	TESS archive (MAST)	Free
Software	100% open-source stack	Free
Compute	Kaggle / Colab GPU, or short cloud-GPU rental	Rs 0 - 8,000

TOTAL

~ **Rs 0**

to a few thousand

Low-risk and deployable: open data, open tools, runs on free-tier GPUs - the cost is engineering effort, not capital.

APEX-ORACLE

Detect. Explain. Trust.

Finding new worlds in the noise - and proving every discovery.

Bharatiya Antariksh Hackathon 2026 . [YOUR NAME] . [YOUR TEAM NAME]